Master's Thesis

Microlocal Morse Theory and Truncated Microsupport
（超局所モース理論と打切り超局所台）

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2026.3

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1 Introduction

In this paper, We study microlocal Morse theory, which is presented, for instance, in [GKS12]. However, we replace the notion of microsupports with that of truncatd microsupprts, introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KFS03], in course of the study of solutions of boundary value problems of $\mathcal{D}$-modules. It was in the context of such kinds of analytic problems that they studied truncated microsupports. However, we shall apply them to study some geometrical problems. In particular, we shall first prove a truncated version of the noncharacteristic deformation lemma, that is, if we weaken a hypothesis of the lemma and add a certain topological condition to the family of deformation, we can prove a finer result of it. Then, in turn, we can formulate the celebrated microlocal morse lemma, under the light of truncated microsupport. Note that by using the truncated microsupport, we can gain the information of the unique continuation. More precisely, if k is an integer and we consider the truncated microsupport $\mathrm{SS}_k(F)$ of a bounded complex of sheaves F, we can investigate whether the cohomology sheaves $H^j(F)$ propagate or not for $j < k$, and in particular, whether the continuation $H^k(F)$ is unique or not, if it can be extended.

This paper is organized as follows. In Section 2, we review the notations and results from [KS90]. In Section 4, we define truncated microsupport of sheaves, due to [KFS03]. In Section 4, we prove a truncated version of the noncharacteristic deformation lemma. Finally in Section 5, we apply the result in Section 4 to the microlocal Morse theory.

2 Background Material

In this section, we recall some definitions and results from [KS90], following its notations with slight modifications.

We consider a commutative unital ring $\mathbf{k}$ of finite global dimension (e.g. $\mathbf{Z}$ or a field). Let X be a topological space. We denote by $\mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$ (resp. $\mathsf{D}^{+}(\mathbf{k}_X)$) the bounded derived category (resp. bounded from below derived category) of sheaves of $\mathbf{k}$-modules on X.

Let us denote by Z a locally closed set of X and by $i\colon Z \hookrightarrow X$ an inclusion. We use the following functors:

$$\mathrm{R}\Gamma_Z(\,\cdot\,) \coloneqq \mathrm{R}i_* i^!, \quad (\,\cdot\,)_Z \coloneqq \mathrm{R}i_! i^{-1}.$$

If there is no risk of confusion, we will write $\mathbf{k}_Z$ rather than $\mathbf{k}_{X,Z}$.

Geometrical Notions

Let M be a real manifold of class C^∞. We denote by $\pi_M\colon T^*M \to M$ the cotangent bundle of M. If there is no risk of confusion, we will write π rather than π_M. If $f\colon M \to N$ is a morphism of manifolds, then we have the natural diagram.

$$\begin{tikzcd}
TM \arrow[r, "f'"] \arrow[dr, "\tau_M"'] & M \underset{N}{\times} TN \arrow[r, "f_\tau"] \arrow[d] \arrow[dr, phantom, "\square"] & TN \arrow[d, "\tau_N"] \\
& M \arrow[r, "f"'] & N,
\end{tikzcd}
\qquad
\begin{tikzcd}
T^*M \arrow[dr, "\pi_M"'] & M \underset{N}{\times} T^*N \arrow[l, "f_d"'] \arrow[r, "f_\pi"] \arrow[d] \arrow[dr, phantom, "\square"] & T^*N \arrow[d, "\pi_N"] \\
& M \arrow[r, "f"'] & N.
\end{tikzcd}$$

In this diagram, f_τ and f_π denote the natural projections. f' is defined by $(x; v) \mapsto (x; df_x v)$, and f_d is induced by the transposition of f'.

Approximation by Inductive Limits

Proposition 2.1 ([KS90, Proposition 2.5.1])**.** Let X be a topological space, Z a subspace, F a sheaf on X. Consider the canonical morhism:

$$\psi\colon \varinjlim_{U} \Gamma(U; F) \to \Gamma(Z; F),$$

where U ranges through the family of open neighborhoods of Z in X.

(i) The morphism ψ is injective.

(ii) Assume that X is Hausdorff and Z is compact. Then ψ is an isomorphism.

(iii) Assume that X is paracompact and Z is closed. Then ψ is an isomorphism.

A paracomact space is by definition a Hausdorff space.

The Mittag-Leffler Condition

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary.

For a projective system of abelian groups $(M_n)_n$, we set

$$M_\infty := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$E_n^\bullet \colon \cdots \to M_n^{j-1} \to M_n^j \to M_n^{j+1} \to \cdots , \tag{2.1}$$

and projective limit

$$E_\infty^\bullet \colon \cdots \to M_\infty^{j-1} \to M_\infty^j \to M_\infty^{j+1} \to \cdots . \tag{2.2}$$

Denote by

$$\varPhi_k \colon H^k(E_\infty^\bullet) \to \varprojlim_n H^k(E_n^\bullet) \tag{2.3}$$

the natural morphism.

Proposition 2.2 ([KS90, Proposition 1.12.4])**.** Assume that for each $k \in \mathbf{Z}$, the system $\left(M_n^{k-1}\right)_n$ satisfies the ML condition. Then

(a) for each $k \in \mathbf{Z}$, the map $\varPhi_k$ is surjective,
(b) if moreover, for a given i the system $\left(H^{i-1}(E_n^\bullet)\right)_n$ satisfies the ML condition, then $\varPhi_i$ is bijective.

The following is called Kashiwara's lemma.

Proposition 2.3 ([KS90, Proposition 1.12.6])**.** Let $(X_s, \rho_{s,t} \colon X_t \to X_s)_{(s,(s \leqq t))}$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps

$$\lambda_s \colon X_s \to \varprojlim_{r<s} X_r$$

and

$$\mu_s \colon \varinjlim_{t>s} X_t \to X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0,s_1} $(s_0 \leqq s_1)$ are injective (resp. surjective).

ML procedure for sheaves

Proposition 2.4 ([KS90, Proposition 2.7.1])**.** Let X be a topological space and let $F \in \mathsf{D}^+(\mathbf{Z}_X)$. Let $(U_n)_{n\in\mathbf{N}}$ be an increasing sequence of open subsets of X, $(Z_n)_{n\in\mathbf{N}}$ a decreasing sequence of closed subsets of X. Set $U := \bigcup_n U_n$, $Z := \bigcap_n Z_n$.

(i) For any j, the natural map $\varphi_j\colon H^j_Z(U;F) \to \varprojlim_n H^j_{Z_n}(U_n;F)$ is surjective.

(ii) Assume that for a given j, the projective system $(H^{j-1}_{Z_n}(U_n;F))_n$ satisfies the ML condition. Then φ_j is bijective.

(iii) Let now $(X_n)_{n\in\mathbf{N}}$ be an increasing family of subsets of X satisfying $X = \bigcup_n X_n$ and $X_n \subset \mathrm{Int}(X_{n+1})$ for all n. Assume that for a given j, the projective system $\bigl(H^{j-1}(X_n;F)\bigr)_n$ satisfies the ML condition. Then the natural map $H^j(X;F) \to \varprojlim_n H^j(X_n;F)$ is bijective.

3 Truncated Microsupport

We give in this section the definition of a truncated microsupport and its functorial property under direct images of proper morphisms.

Definition 3.1. Let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$. The closed conic subset $\mathrm{SS}_k(F)$ of T^*X is defined by: $p \notin \mathrm{SS}_k(F)$ if and only if there exists an open conic neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1-function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

$$H^j_{\{\varphi \geqq 0\}}(F)_x = 0 \quad \text{for any } j \leqq k. \tag{3.1}$$

A Functorial Property under Proper Direct Images

Proposition 3.2. Let $f\colon X \to Y$ be a morphism of manifolds and let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$ such that f is proper on $\mathrm{supp}(F)$. Then for any $k \in \mathbf{Z}$,

$$\mathrm{SS}_k(\mathrm{R}f_*F) \subset f_\pi f_d^{-1}(\mathrm{SS}_k(F)). \tag{3.2}$$

The equality holds in case f is a closed embedding.

4 Truncated Noncharacteristic Deformation Lemma

In this section, we prove a truncated version of the noncharacteristic deformation lemma. First we state the result. Because the original proof of the lemma cannot be applied to our truncatd version, we make an additional assumption of the family of deformation. Therefore, to make this assumption acceptable to the reader, we give several examples of families of deformation before the proof.

Proposition 4.1 (Truncated Noncharacteristic Deformation Lemma)**.** Let X be a Hausdorff space. Let $F \in \mathsf{D}^{+}(\mathbf{k}_X)$, and $(U_t)_{t\in\mathbf{R}}$ be a family of open subsets of X. Let $k \in \mathbf{Z}$. If s is a real number, set $Z_s \coloneqq \bigcap_{t>s} \overline{U_t - U_s}$. We assume the following conditions.

(i) $U_t = \bigcup_{s<t} U_s$, for all $t \in \mathbf{R}$.

(ii) For all pairs (s,t) with $s \leqq t$, the set $\overline{U_t - U_s} \cap \mathrm{supp}\, F$ is compact.

(iii) For all $s \in \mathbf{R}$ and all $x \in Z_s - U_s$, we have

$$(H^j_{X-U_s}(F))_x = 0 \quad \text{for all } j \leqq k.$$

(iv) For all pairs (s,t) with $s<t$, $Z_s \subset U_t$.

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$H^j\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \xrightarrow{\sim} H^j\left(U_t; F\right) \quad \text{for all } j<k,$$

and the injections

$$H^k\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \hookrightarrow H^k\left(U_t; F\right).$$

Example 4.2. The condition (iv) of Proposition 4.1 is not contained in the original version ([KS90, Proposition 2.7.2]). Therefore let us give an example of $(U_t)_t$ that does not satisfy the condition (iv).

(i) Let $X=\mathbf{R}^2$, and for a real number t define the function

$$\varphi_t(x) \coloneqq \begin{cases} \max\left(0, t(2-|x|)\right) & (|x|<2), \\ 0 & (|x| \geqq 2). \end{cases}$$

Then let us denote by U_t the open subset of X

$$U_t \coloneqq \{(x,y)\in X;\ y<\varphi_t(x)\}.$$

This family $(U_t)_t$ does not satisfy the condition (iv). Indeed, if $s<t$, we have

$$Z_s = \{(x,y)\in X;\ y=\varphi_s(x)\}$$

and the point $(2,0)\in Z_s$ is not contained in U_t. Moreover, let us denote by Z the locally closed subset

$$Z \coloneqq \left\{(x,y)\in X;\ (x+2)^2+(y-1)^2=1\right\} \setminus \{(-1,0)\}.$$

Then we denote by $\mathbf{k}_Z$ the sheaf on X. If $j \leqq 0$, we have $H^j_{X\smallsetminus U_t}(\mathbf{k}_Z)_x = 0$ for all $t>s$ and $x\in Z_s \smallsetminus U_t$, but the morhism $H^0(U_1;\mathbf{k}_Z)=\mathbf{k}\to 0 = H^0(U_0;\mathbf{k}_Z)$ is not injective. This give an example that satisfies the conditions (i)–(iii) but not (iv), which result in that the consequence of the proposition does not hold. Hence the condition (iv) is necessary.

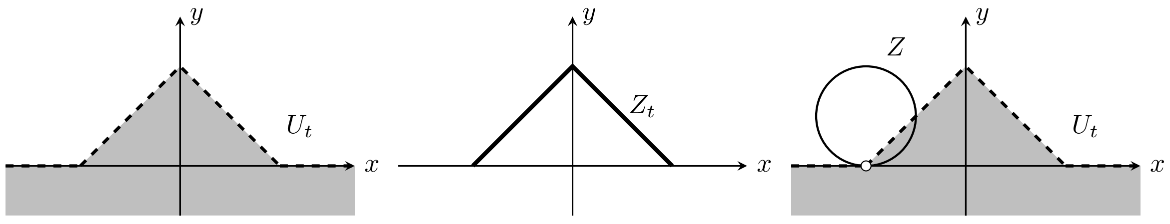

Fig.1 example (i)

Let us, in turn, give three examples of $(U_t)_t$ satisfying the condition (iv). In all the examples below, X is $\mathbf{R}^2$.

(ii) For a real number t, define the open set U_t by

$$U_t := \begin{cases} \{(x,y) \in X;\ \|(x,y)\| < t\} & (t > 0), \\ \varnothing & (t \leqq 0). \end{cases}$$

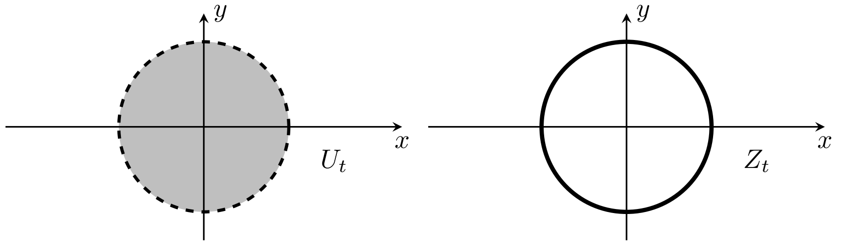

Fig.2　example (ii)

(iii) For a real number t, define the function $\varphi_t \colon \mathbf{R} \to \mathbf{R}$ as follows. If $t \leqq 0$, set $\varphi_t(x) \equiv 0$. If $t > 0$, set

$$\varphi_t(x) := \begin{cases} \sqrt{t^2 - x^2} & (|x| < s), \\ 0 & (|x| \geqq s). \end{cases}$$

Then define the open set U_t by

$$U_t := \{(x,y) \in X;\ y < \varphi_t(x)\}.$$

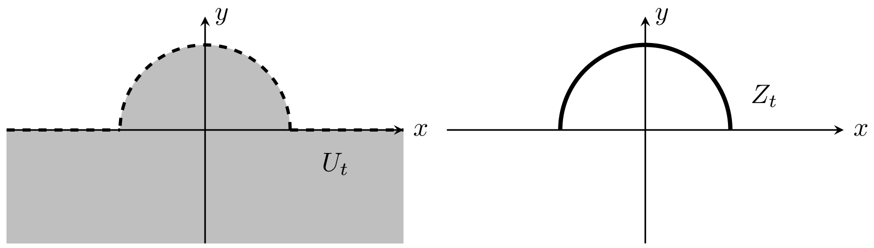

Fig.3　example (iii)

(iv) For a real number t, define the open set U_t as follows. If $t \leqq 0$, set

$$U_t := \{(x,y) \in X;\ y < 0\}.$$

If $t > 0$, set

$$U_t := \left\{ (x,y) \in X;\ \begin{matrix} \text{If } |x| < \frac{1}{t}, & y < t. \\ \text{Otherwise,} & y < \frac{1}{|x|}. \end{matrix} \right\}.$$

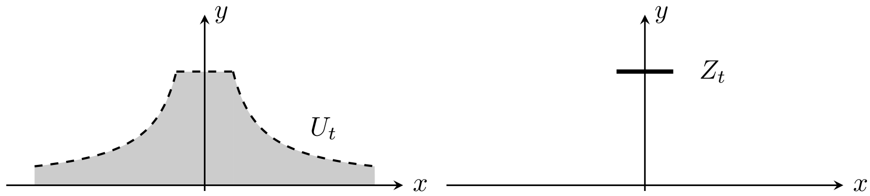

Fig.4　example (iv)

Proof. Consider the following conditions for $j \in \mathbf{Z}$ and $s \in \mathbf{R}$:

$(a)_s^j$ $$\mu_s^j\colon \varinjlim_{t>s} H^j(U_t;F) \xrightarrow{\sim} H^j(U_s;F),$$

$(a')_s^k$ $$\mu_s^k\colon \varinjlim_{t>s} H^k(U_t;F) \hookrightarrow H^k(U_s;F),$$

$(b)_s^j$ $$\lambda_s^j\colon H^j(U_s;F) \xrightarrow{\sim} \varprojlim_{r<s} H^j(U_r;F).$$

Let us prove, for all $s \in \mathbf{R}$, $(a)_s^j$ for $j<k$ and $(a')_s^k$. We may assume that $\overline{U_t - U_s}$ is compact for any $s \leqq t$ by replacing X by $\operatorname{supp} F$. Consider the distinguished triangle

$$\mathrm{R}\Gamma_{X-U_s}(F) \to F \to \mathrm{R}\Gamma_{U_s}(F) \xrightarrow{+1} .$$

Let us endow with $Z_s \cup U_s$ the induced topology, and denote by $i_s\colon Z_s \cup U_s \hookrightarrow X$ the natural continuous map. Applying the functor

$$(\ \cdot\ )|_{Z_s\cup U_s} = i_s^{-1}$$

to the distiguished triangle above, we have

$$\mathrm{R}\Gamma_{X-U_s}(F)|_{Z_s\cup U_s} \to F|_{Z_s\cup U_s} \to \mathrm{R}\Gamma_{U_s}(F)|_{Z_s\cup U_s} \xrightarrow{+1} .$$

Again applying to this triangle the functor $\mathrm{R}\Gamma(Z_s\cup U_s;\ \cdot\ )$, we have

(4.1) $$\mathrm{R}\Gamma(Z_s\cup U_s;\mathrm{R}\Gamma_{X-U_s}(F)) \to \mathrm{R}\Gamma(Z_s\cup U_s;Z_s\cup U_s) \to \mathrm{R}\Gamma(U_s;F) \xrightarrow{+1} .$$

By the hypothesis (iii) and the definition of a local cohomology, we see that

$$H^j_{X-U_s}(F)\Big|_{Z_s\cup U_s} = 0 \quad \text{for any } j \leqq k,$$

and then

$$H^j\left(Z_s\cup U_s;\mathrm{R}\Gamma_{X-U_s}(F)\right) = 0 \quad \text{for any } j \leqq k.$$

Thus we have

(4.2) $$H^j(Z_s\cup U_s;F) \xrightarrow{\sim} H^j(U_s;F) \quad \text{for any } j<k,$$

(4.3) $$H^k(Z_s\cup U_s;F) \hookrightarrow H^k(U_s;F).$$

Therefore, if we show Lemma 4.3 below, we have

$$\begin{aligned}\varinjlim_{t>s} H^j(U_t;F) &\cong H^j(Z_s\cup U_s;F)\\ &\xrightarrow{\sim} H^j(U_s;F)\end{aligned}$$

for any $j < k$, and $(a)_s^j$ follows. Similarly, we have

$$\begin{aligned}\varinjlim_{t>s} H^k(U_t;F) &\cong H^k(Z_s \cup U_s;F)\\ &\hookrightarrow H^k(U_s;F)\end{aligned}$$

and $(a')_s^k$ follows.

Lemma 4.3. For any $j \in \mathbf{Z}$ and $s \in \mathbf{R}$, there exists a natural isomorphism

$$\varinjlim_{t>s} H^j(U_t;F) \cong H^j(Z_s \cup U_s;F).$$

Proof of Lemma 4.3 First we prove that for $F \in \mathrm{Mod}(\mathbf{k}_X)$, the natural morphism

$$\varinjlim_{t>s} \Gamma(U_t;F) \xrightarrow{\sim} \Gamma(Z_s \cup U_s;F). \tag{4.4}$$

is isomorphic. Consider the following diagram.

$$\begin{tikzcd}
\varinjlim_{t>s} \Gamma(U_t;F) \arrow[d, "\tilde{\phi}"'] \arrow[dr] & \\
\varinjlim_{U\supset Z_s} \Gamma(U \cup U_s;F) \arrow[r, "\phi"'] \arrow[d] & \Gamma(Z_s \cup U_s; F|_{Z_s\cup U_s}) \arrow[d] \\
\varinjlim_{U\supset Z_s} \Gamma(U;F) \arrow[r, "\psi"'] & \Gamma(Z_s; F|_{Z_s})
\end{tikzcd}$$

Let us begin by showing that the morphism

$$\tilde{\phi} \colon \varinjlim_{t>s} \Gamma(U_t;F) \to \varinjlim_{U\supset Z_s} \Gamma(U \cup U_s;F)$$

is an isomorphism. To prove this, it is enough to show that $(U_t)_{t>s}$ is a cofinal family of $(U \cup U_s)_{U\supset Z_s}$. Let U be an open neighborhood of Z_s. Then we shall show by contradiction that there exists a real number $t > s$ such that $U_t \subset U \cup U_s$. Let us assume that $U_t \setminus (U \cup U_s) \neq \varnothing$ for any $t > s$. Then define the family of subsets $(A_r)_{s<r<t}$ of $\overline{U_t \setminus U_s}$ by $A_r := U_r \setminus (U \cup U_s)$. This $(A_r)_{s<r<t}$ satisfies the finite intersection property. Indeed, for finite real numbers $s < r_0 < \cdots < r_m < t$, it follows from the hypothesis that

$$\begin{aligned}A_{r_0} \cap \cdots \cap A_{r_m} &= A_{r_0}\\ &= U_{r_0} \setminus (U \cup U_s)\\ &\neq \varnothing.\end{aligned}$$

Therefore, since $\overline{U_t \setminus U_s}$ is compact, we have $\bigcap_{s<r<t} \overline{A_r} \neq \varnothing$. However, we have from the hypothesis

$Z_s \subset U$ that

$$\begin{aligned}
\varnothing \neq \bigcap_{s<r<t} \overline{A_r} &= \bigcap_{s<r<t} \overline{U_r \smallsetminus (U \cup U_s)} \\
&= \bigcap_{s<r<t} (\overline{U_r \smallsetminus U_s} \cap \overline{U_r \smallsetminus U}) \\
&= \bigcap_{s<r<t} \overline{U_r \smallsetminus U_s} \cap \bigcap_{s<r<t} \overline{U_r \smallsetminus U} \\
&= Z_s \cap \bigcap_{s<r<t} \overline{U_r \smallsetminus U} \\
&= Z_s \cap \bigcap_{s<r<t} \overline{U_r} \cap \overline{X \smallsetminus U} \\
&= Z_s \cap \bigcap_{s<r<t} \overline{U_r} \cap (X \smallsetminus U) \\
&= Z_s \cap (X \smallsetminus U) \cap \bigcap_{s<r<t} \overline{U_r} \\
&= \varnothing.
\end{aligned}$$

This is a contradiction, and it follows that $\tilde{\phi}$ is an isomorphism.

By this isomorphism, it is enough to show that

$$\phi\colon \varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F) \to \Gamma(Z_s \cup U_s; F)$$

is isomorphic. Since X is Hausdorff and Z_s is compact, the natural morhism

$$\psi\colon \varinjlim_{U \supset Z_s} \Gamma(U; F) \to \Gamma(Z_s; F)$$

is isomorphic by Proposition 2.1. Since U_s is open, we have another isomorphism

$$\psi'\colon \varinjlim_{V \supset U_s} \Gamma(V; F) \to \Gamma(U_s; F).$$

The injectivity of ϕ follows from the definition of a sheaf and the injectivities of ψ and ψ'.

Let us prove that ϕ is surjective. Let $\sigma \in \Gamma(Z_s; F)$ be a section of $F|_{Z_s \cup U_s}$. By the isomorphism ψ above, we can extend $\sigma|_{Z_s}$ to an open neighborhood U of Z_s in X. More precisely, there exists a section τ on U such that $\tau|_{Z_s} = \sigma|_{Z_s}$. Now we see that $\tau|_{Z_s} = (\tau|_{(Z_s \cup U_s) \cap U})\Big|_{Z_s}$, and by the definition of a section on the closed subset Z_s of $(Z_s \cup U_s) \cap U$, we can find an open subset W of $(Z_s \cup U_s) \cap U$ such that

$$\tau|_W = \sigma|_W. \tag{4.5}$$

By the definition of the induced topology of $(Z_s \cup U_s) \cap U$, there exists an open set $\widetilde{W}$ of X such that

$$W = ((Z_s \cup U_s) \cap U) \cap \widetilde{W} \quad \text{and} \quad \widetilde{W} \subset U.$$

Thus we have $W = ((Z_s \cup U_s)) \cap \widetilde{W}$. For this $\widetilde{W}$, the pair $\left(Z_s \cup U_s, \widetilde{W}\right)$ is an open covering of $\widetilde{W} \cup U_s$. Thus by (4.5), we can glue σ and $\tau|_{\widetilde{W}}$ together and get a section $\widetilde{\sigma}$ such that

$$\widetilde{\sigma}|_{Z_s \cup U_s} = \sigma, \quad \widetilde{\sigma}|_{\widetilde{W}} = \tau|_{\widetilde{W}}.$$

This proves the surjectivity of ϕ.

Next Let $F \in \mathsf{D}^+(\mathbf{k}_X)$. Let $I^\bullet$ be a complex of flabby sheaves quasi-isomorphic to F. Then it follows from (4.4) that

$$\begin{aligned}
\varinjlim_{t>s} H^j(U_t; F) &\cong \varinjlim_{t>s} H^j\Gamma(U_t; I^\bullet) \\
&\cong H^j\left(\varinjlim_{t>s} \Gamma(U_t; I^\bullet)\right) \\
&\cong H^j\Gamma(Z_s \cup U_s; I^\bullet) \\
&\cong H^j(Z_s \cup U_s; F).
\end{aligned}$$

This completes the proof. □

Let us come back to the proof of Proposition 4.1. We prove $(b)^j_s$ for all $j \in \mathbf{Z}$ and $s \in \mathbf{R}$ by induction on j. Since $F \in \mathsf{D}^+(\mathbf{k}_X)$, we have $(b)^j_s$ for $j \ll 0$ and $s \in \mathbf{R}$. Hence there exists an integer $j_0 < k$ such that $(b)^j_s$ holds for any $j < j_0$ and $s \in \mathbf{R}$. Then assume $(b)^j_s$ is proved for all $j < j_0$ and $s \in \mathbf{R}$. Since we have $(a)^j_s$ and $(b)^j_s$ for each $j < j_0$ and each $s \in \mathbf{R}$, we can apply Kashiwara's lemma (Proposition 2.3) and have

$$H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F) \quad \text{for } j < j_0 \text{ and } s \leqq t.$$

Fix $s \in \mathbf{R}$ and consider the projective system $\left(H^{j_0-1}(U_{s-1/n})\right)_{n\in\mathbf{N}}$. This system satisfies ML condition. Therefore, by applying Proposition 2.4 (iii), we have

$$H^j(U_s; F) \xrightarrow{\sim} \varprojlim_n H^j\left(U_{s-\frac{1}{n}}; F\right).$$

Since $(s - 1/n)_n$ is a cofinal family of $(t)_{s>t}$, we have $(b)^{j_0}_s$. Then by induction on j, we have $(b)^j_s$ for $j < k$ and $s \in \mathbf{R}$. Again by applying Proposition 2.4 to the projective system $\left(H^j(U_n; F)\right)_n$, we obtain

$$\begin{aligned}
H^j\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) &\cong H^j\left(\bigcup_{n\in\mathbf{N}} U_n; F\right) \\
&\xrightarrow{\sim} \varprojlim_n H^j\left(U_n; F\right)
\end{aligned}$$

and for each n, we have by Kashiwara's lemma that

$$H^j\left(U_n; F\right) \cong H^j(U_s; F)$$

for $n \geqq s$.

Finally we replace $(a)^j_s$ by $(a')^k_s$ for $j = k$, and obtain

$$H^k\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \hookrightarrow H^j\left(U_s; F\right).$$

This completes the proof. □

5 An Application to Microlocal Morse Theory

In this section, we shall apply the result in the previous section to a particular case of a truncated version of microlocal Morse lemma.

Let M be a manifold of class C^∞. If $t \in \mathbf{R}$, we denote the open interval $(-\infty, t)$ by I_t.

Lemma 5.1. Let $-\infty < a < b < +\infty$ be real numbers and $G \in \mathsf{D}^{\mathrm{b}}(\mathbf{R})$. Let k be an integer. If

$$\left(H^j_{[t,+\infty)}(G)\right)_t \cong 0 \quad \text{for all } t \in [a,b)\text{ , and all } j \leqq k.$$

Then one has

$$H^j((-\infty,b)\,;G) \xrightarrow{\sim} H^j((-\infty,a)\,;G)$$

for all $j < k$, and this morphism is injective for $j = k$.

Proof. We will use Proposition 4.1. Let us put $X = (-\infty, b)$, $F = G|_{I_b}$, and $U_t = I_t$, and verify the hypotheses of the proposition.

(i) $\bigcup_{s<t} I_s = \bigcup_{s<t}(-\infty, s) = (-\infty, t) = I_t$ for all $t \in \mathbf{R}$.

(ii) If $s \leqq t$, then $\overline{I_t - I_s} \cap \operatorname{supp} G = \overline{[s,t)} \cap \operatorname{supp} G = [s,t] \cap \operatorname{supp} G$ is compact.

(iii) For $s \in \mathbf{R}$, we have

$$Z_s = \bigcap_{t>s} \overline{I_t - I_s} = \bigcap_{t>s} \overline{[s,t)} = \bigcap_{t>s} [s,t] = \{s\}\,,$$

and

$$Z_s - I_s = \{s\}\,.$$

By the hypothesis, we have

$$H^j_{X-I_s}(F)_s = H^j_{[s,\infty)}(F)_s = 0$$

for $s \in \{s\}$.

(iv) It follows that $Z_s = \{s\} \subset (-\infty, t) = I_t$ if $s < t$.

Then we can apply Proposition 4.1, and we have the isomorphisms for all $t \in [a,b)$ and $j < k$,

$$H^j(I_b; F) = H^j\left(\bigcup_s I_s; F\right) \xrightarrow{\sim} H^j(I_t; F)$$

and the injections

$$H^k\left(I_b; F\right) = H^k\left(\bigcup_s I_s; F\right) \hookrightarrow H^k(I_t; F).$$

□

Theorem 5.2. Let F in $\mathsf{D}^{\mathrm{b}}(\mathbf{k}_M)$, and $\psi\colon M \to \mathbf{R}$ be a real-valued function of class C^1, proper on $\operatorname{supp}(F)$. Let $a < b$ be real numbers. We assume $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leqq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t \coloneqq \psi^{-1}\left((-\infty, t)\right)$. Then, the restriction morphisms

$$H^j(M_b; F) \to H^j(M_a; F)$$

are isomorphic for each $j < k$, and injective for $j = k$.

Proof. First, we have

$$H^j(M_t;F) \cong H^j\,\mathrm{R}\Gamma((-\infty,t)\,;\mathrm{R}\psi_*F)$$

for $t \in I$. Since we assume that $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leqq \psi(x) < b$, and by (3.2), the formula $\mathrm{SS}_k(\mathrm{R}\psi_*F) \subset \psi_\pi \psi_d^{-1}\,\mathrm{SS}_k(F)$ holds. Hence we have

$$\left(H^j_{[\psi(x),+\infty)}(\mathrm{R}\psi_*F)\right)_{\psi(x)} = 0$$

for each $j \leqq k$. Then we can apply Lemma 5.1 and have

$$H^j((-\infty,b)\,;\mathrm{R}\psi_*F) \xrightarrow{\sim} H^j((-\infty,a)\,;\mathrm{R}\psi_*F)$$

for all $j < k$ and

$$H^k((-\infty,b)\,;\mathrm{R}\psi_*F) \hookrightarrow H^k((-\infty,a)\,;\mathrm{R}\psi_*F).$$

Then we can conclude that

$$H^j(M_b\,;F) \xrightarrow{\sim} H^j(M_a\,;F)$$

for all $j < k$ and

$$H^k(M_b\,;F) \hookrightarrow H^k(M_a\,;F).$$

□

Acknowledgement

We would like to thank Naofumi Honda, our superviser for teaching us all the important things about mathematics. We also thank Ryosuke Sakamoto, Kango Ito, Yoshiaki Yamamura for fruitiful discussions. In particular, Sakamoto-san gave us several ideas that is contained in this paper. We would like to thank Yuichi Ike for giving us some fundamental ideas of this paper.

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